

Protik Mukherjee

Toronto, Canada

protik.mkj@gmail.com | [LinkedIn](#)

RESEARCH INTERESTS

Machine Learning for IoT/CPS, Federated and Distributed Learning, Digital Twins and Model-Driven Engineering, Edge AI, Multimodal Signal Processing and Health Informatics.

EDUCATION

Toronto Metropolitan University (TMU), Toronto, Canada Sep 2023 – Sep 2024
Master of Engineering in Electrical and Computer Engineering Cumulative GPA: 4.10/4.33
Specialization: Artificial Intelligence

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh Feb 2018 – May 2022
Bachelor of Science in Electrical, Electronic and Communication Engineering Cumulative GPA: 3.36/4.00

ACADEMIC EXPERIENCE

Software Modelling Lab (SML), Toronto Metropolitan University, Toronto, Canada January 2025 – Present
Canada
Research Assistant

- Designed a statechart-driven Smart Home Digital Twin integrating four ESP32 subsystems with real-time Firebase synchronization, presented at EDTConf 2025.
- Co-developed ArchML4IoT, a domain-specific architecture modelling language and toolchain that transforms high-level IoT architectural specifications into executable statecharts and configuration artifacts, accepted at ANNSIM 2026.
- Developed rolling 10-minute ML pipelines for fire risk, occupancy, and power prediction using Random Forests and LSTMs with cloud-based alerting.
- Analyzed latency and noise robustness and produced LaTeX-based technical documentation and evaluation reports.

Campbell Lab, Department of Chemistry and Biology, Toronto Metropolitan University, Toronto, Canada Mar 2026 – Present
Research Assistant

- Supporting sensor-based crop monitoring and data analysis for indoor horticulture research focused on year-round berry production in controlled environments.
- Installing, calibrating, and maintaining sensing equipment for controlled-environment agricultural experiments.

Finance Department, Ted Rogers School of Management, Toronto Metropolitan University, Toronto, Canada Jun 2026 – Present
Research Assistant

- Built a daily price and volume dataset of 1,053 cryptocurrencies covering 2014 to 2026 (about 1.29 million rows) in Python with pandas, using Yahoo Finance and CoinGecko as sources.
- Cleaned and checked the dataset by removing duplicate rows caused by shared ticker symbols, blanking invalid prices with a written log, and comparing sample days against the source.
- Calculated summary statistics, monthly and quarterly returns, volatility and pairwise correlations for five financial assets, and prepared the tables and charts in Python and formula-based Excel workbooks.

WINCORE Lab, Toronto Metropolitan University, Toronto, Canada Mar 2024 – December 2024
Research Assistant

- Implemented federated anomaly detection for smart-city IoT systems using FedProx and FedAvg with transformer, autoencoder backbones, and achieved 86.43% accuracy on non-IID data.
- Benchmarked communication rounds and convergence stability across heterogeneous clients using PyTorch.

Toronto Metropolitan University, Toronto, Canada

Sep 2024 – Present

Head Teaching Assistant and Teaching Assistant

- ELE792 (Digital Signal Processing): Head TA for the Fall 2026 hardware labs on Texas Instruments TMS320C6713 DSK boards in Code Composer Studio. Prepared the Lab 1 reference solution and lab runbook.
- ELE754 (Power Electronics): Supporting weekly two-hour labs in the power electronics lab in Fall 2026.
- DG8114 (Human-Robot Interaction): Supervised projects on telepresence and robotic interface systems.
- COE608 (Computer Organization and Architecture): Guided FPGA/VHDL CPU design labs and project evaluations.
- CPS188 (Computer Programming Fundamentals): Mentored students on C programming and logical problem solving.
- Also supported courses in field theory, electronic circuits, electromagnetics, electric networks, and communication studies (ELE401, ELE404, ELE531, ELE302, CMN216).

University of Toronto Mississauga (UTM), Mississauga, Canada

Sep 2024 – Apr 2026

Instructional Assistant

- CCT285H5S (Immersive Environment Design): Supported VR/AR design projects, assessed iterative prototypes, and guided students on interaction flow and usability.
- CCT461H5S (Speculative Design III): Evaluated VR/AR design projects and advised on interaction design using emerging technologies.
- Also supported courses in interactive media design, mobile app prototyping, and advanced web development (CCT286, CCT361, CCT460).

PUBLICATIONS

Peer-Reviewed Papers

- Mukherjee, P., Khwaja, A. S., Anpalagan, A., Baljon, M. (2025). Federated Learning for Anomaly Detection in Smart Cities. 2025 International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET), IEEE.
- Ovi, T. B., Bashree, N., Mukherjee, P., Parthima, M. A. (2023). RoadDuAttNet: A Dual-Attention Based U-Net Architecture for Road Extraction from Satellite Image Utilizing Regional Features. ICICT4SD 2023, IEEE.
- Ovi, T. B., Bashree, N., Mukherjee, P., Mosharrof, S., Parthima, M. A. (2023). Performance Analysis of Various EfficientNet-Based U-Net++ Architecture for Automatic Building Extraction from High-Resolution Satellite Images. In: Human-Centric Smart Computing (ICHCSC 2023), Smart Innovation, Systems and Technologies (SIST), Vol. 376. Springer, Singapore.
- Salma, S., Zha, Z., Mukherjee, P., Mustafiz, S. (2025). Engineering Digital Twins with Statecharts: A Smart Home Application. IEEE Xplore (EDTConf 2025, MODELS Satellite).

Accepted

- Zha, Z., Mukherjee, P., Mustafiz, S. (2026). Architecture-Based Generation of Simulation Models for IoT Twins. Accepted at the 2026 Annual Modeling and Simulation Conference (ANNSIM'26), Orlando, FL, USA.

PROJECTS

Smart Home Digital Twin with Statecharts (ESP32 + Firebase)

2024 – 2025

Lead Developer

Toronto, Canada

- Developed four-ESP architecture integrating fire, garage, and lighting modules with centralized ML reasoning and Firebase cloud sync.
- Designed OTA-ready ESP-NOW communication and energy monitoring modules for digital twin feedback loops.

Federated Learning for Anomaly Detection in Smart Cities

2024

Research Lead

Toronto, Canada

- Constructed a FedProx-based transformer-autoencoder pipeline for non-IID IoT anomaly detection, optimizing communication rounds and convergence.

Stacked Machine Learning for Chronic Disease Detection

2022

Undergraduate Researcher

Dhaka, Bangladesh

- Built ensemble classifiers combining stacked learning and hybrid voting techniques for early detection of diabetes and heart disease from clinical patient data.
- Evaluated model generalization across imbalanced health datasets using precision-recall analysis and cross-validation.

Smart Meter for Electricity

2018

Team Member

Dhaka, Bangladesh

- Developed ML-assisted smart metering architecture to monitor energy distribution along with digitalized payment system, and implemented early anomaly detection system.

SELECTED COURSES

Master's Courses: Deep Learning, Neural Networks, NLP (Text Mining), Intelligent Systems, Reconfigurable Computing Systems, Digital Systems Testing, Computer Systems Modeling.

Bachelor's Courses: Signals and Systems, Communication Theory, Control Systems, Power Systems, Microprocessors and Interfacing, Digital Signal Processing, Telecommunication Engineering, VLSI Design, Complex Variables and Statistics.

AWARDS

Dean's List, Academic Excellence, Military Institute of Science and Technology 2020

Winner, National Power and Energy Hackathon, Smart Meter for Electricity Project 2018

Divisional Champion, Prothom Alo National Debate Carnival, recognized for public speaking and critical reasoning among 60+ teams 2013

Runner-up, Bangladesh Mathematical Olympiad, national-level recognition for problem-solving in mathematics 2012 & 2014

OTHER PROFESSIONAL EXPERIENCES

British American Tobacco Bangladesh (BATB), Dhaka, Bangladesh Jun 2023 – Aug 2023

Electronic Executive, Legal and External Affairs

- Managed RFID access for 800+ staff and 50+ zones and streamlined credential processing to 24-hour turnaround.
- Monitored a 200+ camera CCTV network ensuring 99.9% uptime and reducing incident response by 30%.
- Oversaw UPS and server room power systems achieving zero unplanned outages during tenure.

Robi Axiata Limited, Dhaka, Bangladesh Dec 2022 – Apr 2023

Engineering Intern, Infrastructure Management

- Improved Digital Asset Management workflows for network infrastructure and automated material-return process reducing errors by 25%.

IEEE MIST Student Branch, Dhaka, Bangladesh 2021 – 2022

Chair

- Organized workshops, hackathons, and research seminars and fostered student–faculty collaborations.

SKILLS

Programming Languages: Python, C, C++, Java, MATLAB.

Data Analysis & Reporting: Data preprocessing, KPI tracking, Excel dashboards, PowerPoint reporting.

ML & AI Tools: Model development, TensorFlow, PyTorch, Federated learning, Time-series and sensor signal analysis, scikit-learn.

Circuit Design & Embedded Systems: Circuit analysis, FPGA-based processor design (VHDL), microcontroller programming, Digital logic design, hardware–software integration.

Cloud & Deployment: ML model deployment on AWS & GCP, Docker containerization.

Developer Tools: Git, Jupyter Notebook, Visual Studio Code, MIT App Inventor, Thinkable.